

Grade 11 Formula Sheet

Kinematics

$$\overrightarrow{\Delta d} = \left(\frac{\overrightarrow{v_i} + \overrightarrow{v_f}}{2}\right)\Delta t$$

$$\vec{a} = \frac{\overrightarrow{v_f} - \overrightarrow{v_i}}{\Delta t}$$

$$\overrightarrow{\Delta d} = \overrightarrow{v_i}\Delta t + \frac{1}{2}\vec{a}\Delta t^2$$

$$\overrightarrow{\Delta d} = \overrightarrow{v_f}\Delta t - \frac{1}{2}\vec{a}\Delta t^2$$

$$v_f^2 = v_i^2 + 2a\Delta d$$

Energy

$$KE = \frac{1}{2}mv^2$$

$$E = mc^2$$

$$PE = mgh$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$E_i + W = E_f$$

$$m_p = 1.007276 \text{ u}$$

$$W = Fd\cos(\theta)$$

$$m_n = 1.008665 \text{ u}$$

$$P = \frac{W}{\Delta t}$$

$$m_e = 0.000549 \text{ u}$$

$$\%e = \frac{E_{out}}{E_{in}} \times 100\%$$

$$N = N_0 \left(\frac{1}{2}\right)^{\frac{t}{h}}$$

Dynamics

$$\vec{F}_{net} = m\vec{a}$$

$$F_f = \mu F_N$$

$$F_g = mg$$

$$F_g = \frac{GMm}{r^2}$$

$$G = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$

Electricity

$$Q = Ne$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$I = \frac{Q}{\Delta t}$$

$$V = \frac{E}{Q}$$

$$P = IV = I^2R$$

Circuits

$$I_s = I_1 = I_2 = I_3 \dots$$

$$V_s = V_1 + V_2 + V_3 \dots$$

$$R_s = R_1 + R_2 + R_3 \dots$$

$$I_p = I_1 + I_2 + I_3 \dots$$

$$V_p = V_1 = V_2 = V_3 \dots$$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \dots$$